

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: ARTIFICIAL INTELLIGENCE

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

~~*I **NARAPPAGARI SRAVYA(192472192)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*~~

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

ANNEXURE A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu

II. If a patient has Fever AND Rash $\rightarrow$ Possible Measles

III. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia

IV. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

I. $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]

II. $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$ [Rule 2]

III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]

IV. Facts: $\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance}), \text{Vehicle}(\text{CarA}), \text{Behind}(\text{CarA}, \text{Ambulance})$

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) $P1: \text{SoilDry} \rightarrow \text{IrrigationNeeded}$
- 2) $P2: \text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$
- 3) $P3: \neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$
- 4) Facts: $\text{SoilDry} = \text{True}, \text{CropWheat} = \text{True}$

(a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute).

(b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.

(c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) $\text{Stench}[1,2] \rightarrow \text{Wumpus is adjacent to cell}[1,2]$
- 2) $\text{Breeze}[1,1] \rightarrow \text{Pit is adjacent to cell}[1,1]$
- 3) $\text{Glitter}[x,y] \rightarrow \text{Gold is at}[x,y]$
- 4) $\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding ; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Annexure A

Case Study Solutions — Logical Agents (Propositional & First-Order Logic)

Case Study 1 — Smart Medical Diagnosis System

(a) Propositional Logic Representation

Symbols (Propositional Variables):

Symbol	Meaning
F	Patient has Fever
C	Patient has Cough
R	Patient has Rash
B	Patient has Breathlessness
M1	Possible Flu
M2	Possible Measles
M3	Possible Pneumonia

Rules in Propositional Logic:

Rule 1: $F \wedge C \rightarrow M1$ (Fever AND Cough $\rightarrow$ Possible Flu)

Rule 2: $F \wedge R \rightarrow M2$ (Fever AND Rash $\rightarrow$ Possible Measles)

Rule 3: $C \wedge B \rightarrow M3$ (Cough AND Breathlessness $\rightarrow$ Possible Pneumonia)

Facts for Patient A:

F = True, C = True, B = True (R is not asserted, treated as unknown/False)

(b) Forward Chaining

Forward chaining starts from the known facts and fires any rule whose premises are fully satisfied, adding new derived facts to the knowledge base until no more rules can fire.

Step	Known Facts	Rule Checked	Fires?	Fact Derived
1	F=T, C=T, B=T	Initial facts asserted	—	F, C, B
2	F=T, C=T, B=T	Rule 1: $F \wedge C \rightarrow M1$	Yes (F=T, C=T)	M1 (Possible Flu)
3	F=T, C=T, B=T	Rule 2: $F \wedge R \rightarrow M2$	No (R unknown/False)	— (not derived)
4	F=T, C=T, B=T	Rule 3: $C \wedge B \rightarrow M3$	Yes (C=T, B=T)	M3 (Possible Pneumonia)

Step	Known Facts	Rule Checked	Fires?	Fact Derived
5	F,C,B,M1,M3	No further rule applicable	—	Fixed point reached

Final derived diagnoses for Patient A: Possible Flu (M1) and Possible Pneumonia (M3).

(c) Backward Chaining — Goal: 'Patient A has Pneumonia' (M3)

Backward chaining starts from the goal and recursively reduces it to sub-goals until each sub-goal matches a known fact.

Goal reduction tree:

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M3 (Possible Pneumonia)          [GOAL]
├── Rule 3:  $C \wedge B \rightarrow M3$ 
│   ├── Sub-goal: C (Cough)
│   │   ├── Matches Fact: C = True  ✓ SATISFIED
│   │   └── Sub-goal: B (Breathlessness)
│   │       ├── Matches Fact: B = True  ✓ SATISFIED

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Both sub-goals satisfied $\rightarrow$ Rule 3 fires $\rightarrow M3 = \text{True}$

Conclusion: Since both C and B are true facts, Rule 3 is confirmed, and the goal 'Patient A has Pneumonia' is proved TRUE by backward chaining.

Case Study 2 — Autonomous Traffic Management Agent

(a) Unification of Rule 1 with the Given Facts

Rule 1: $\forall x \text{ Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

Facts: $\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance})$

Step	Unify	Result / Substitution θ
1	$\text{Vehicle}(x)$ with $\text{Vehicle}(\text{Ambulance})$	$\theta = \{x / \text{Ambulance}\}$

Step	Unify	Result / Substitution θ
2	EmergencyType(x) with EmergencyType(Ambulance), applying θ from step 1	EmergencyType(Ambulance) matches directly — consistent with $\theta = \{x /$ Ambulance}
3	Apply θ to conclusion ClearPath(x)	ClearPath(Ambulance)

MGU (Most General Unifier): $\theta = \{x / \text{Ambulance}\}$

Result: Rule 1 fires and derives the new fact ClearPath(Ambulance).

(b) Resolution Proof that Proceed(CarA) is True

Step 1 — Convert all rules to CNF (eliminate implication, standardize variables):

Clause #	Original Rule	CNF Form
C1	Rule 1: $\text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$	$\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$
C2	Rule 2 (conjunct 1): $\text{ClearPath}(x) \rightarrow \text{GreenSignal}(x)$	$\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$
C3	Rule 2 (conjunct 2): $\text{ClearPath}(x) \rightarrow \neg \text{RedSignal}(x)$	$\neg \text{ClearPath}(x) \vee \neg \text{RedSignal}(x)$
C4	Rule 3: $\text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$	$\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x,y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$

Facts (unit clauses): F1: Vehicle(Ambulance) F2: EmergencyType(Ambulance) F3: Vehicle(CarA)
F4: Behind(CarA, Ambulance)

Negated Goal (for refutation): $\neg \text{Proceed}(\text{CarA})$

Step 2 — Resolution steps:

- Resolve C1 {x/Ambulance} with F1 (Vehicle(Ambulance)):
 $\neg \text{EmergencyType}(\text{Ambulance}) \vee \text{ClearPath}(\text{Ambulance}) \quad \dots \text{R1}$
- Resolve R1 with F2 (EmergencyType(Ambulance)):
 $\text{ClearPath}(\text{Ambulance}) \quad \dots \text{R2}$
- Resolve C2 {x/Ambulance} with R2 (ClearPath(Ambulance)):
 $\text{GreenSignal}(\text{Ambulance}) \quad \dots \text{R3}$
- Resolve C4 {x/CarA, y/Ambulance} with F3 (Vehicle(CarA)):
 $\neg \text{Behind}(\text{CarA}, \text{Ambulance}) \vee \neg \text{GreenSignal}(\text{Ambulance}) \vee \text{Proceed}(\text{CarA}) \quad \dots \text{R4}$

5. Resolve R4 with F4 (Behind(CarA,Ambulance)):
 $\neg \text{GreenSignal}(\text{Ambulance}) \vee \text{Proceed}(\text{CarA})$... R5
6. Resolve R5 with R3 (GreenSignal(Ambulance)):
 $\text{Proceed}(\text{CarA})$... R6
7. Resolve R6 with negated goal $\neg \text{Proceed}(\text{CarA})$:
 $\square$ (empty clause — CONTRADICTION)

Since adding the negated goal $\neg \text{Proceed}(\text{CarA})$ to the KB leads to a contradiction (empty clause $\square$), the original goal $\text{Proceed}(\text{CarA})$ is logically entailed by the knowledge base. Hence 'Proceed(CarA)' is proved TRUE.

(c) Sufficiency of the KB for Two Simultaneous Emergency Vehicles

The current KB is NOT sufficient to handle two simultaneous emergency vehicles. Its three rules only reason about a single vehicle's emergency status and one 'Behind' relation at a time; they contain no mechanism to compare, prioritize, or sequence multiple vehicles that simultaneously satisfy $\text{EmergencyType}(x)$. If two emergency vehicles approach the same intersection from conflicting directions, Rule 1 and Rule 2 would independently grant both $\text{ClearPath}(x)$ and $\text{GreenSignal}(x)$, which is physically contradictory (two conflicting green signals at once).

Additional rules/predicates needed, with justification:

- $\text{Priority}(x, n)$ — assigns a numeric priority/severity level to each emergency vehicle (e.g., based on type: ambulance > fire truck > police). Needed so the agent can choose which vehicle receives ClearPath when two both qualify.
- $\text{ConflictsWith}(x, y)$ — a predicate/fact identifying that two routes intersect or share a signal phase. Needed to detect when granting GreenSignal to both x and y would be unsafe.
- Rule: $\text{Vehicle}(x) \wedge \text{Vehicle}(y) \wedge \text{EmergencyType}(x) \wedge \text{EmergencyType}(y) \wedge \text{ConflictsWith}(x,y) \wedge \text{Priority}(x,n1) \wedge \text{Priority}(y,n2) \wedge (n1 > n2) \rightarrow \text{ClearPath}(x) \wedge \neg \text{ClearPath}(y)$. This resolves ties by priority.
- Mutual exclusion constraint: $\neg(\text{GreenSignal}(x) \wedge \text{GreenSignal}(y))$ whenever $\text{ConflictsWith}(x,y)$ — an integrity constraint to prevent the KB from deriving two contradictory green signals.
- Temporal/queueing predicate, e.g. $\text{HoldFor}(y, t)$ — allows the lower-priority emergency vehicle y to be briefly held rather than simply denied, preserving safety while still expediting it once x has cleared.

Without these additions, the KB is logically incomplete for the two-vehicle case: it can derive an inconsistent state (both $\text{ClearPath}(x)$ and $\text{ClearPath}(y)$ true when x and y conflict), which is unacceptable in a safety-critical traffic system.

Case Study 3 — Agricultural Expert Reasoning System

(a) Conversion of P1, P2, P3 and Facts to CNF

P1: $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$

Step 1 (Eliminate $\rightarrow$): $\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

Step 2 ($\neg$ already innermost, no distribution needed)

CNF: $\neg \text{SoilDry} \vee \text{IrrigationNeeded}$... (Clause 1)

P2: $\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$

Step 1 (Eliminate $\rightarrow$): $\neg(\text{IrrigationNeeded} \wedge \text{CropWheat}) \vee \text{ApplyDripMethod}$

Step 2 (Move $\neg$ inward, De Morgan): $\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$

Step 3 (already a disjunction of literals — distribution not required)

CNF: $\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$... (Clause 2)

P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$

Step 1 (Eliminate $\rightarrow$): $\neg(\neg \text{ApplyDripMethod} \wedge \text{CropWheat}) \vee \text{CropAtRisk}$

Step 2 (Move $\neg$ inward, De Morgan): $\text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$

CNF: $\text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$... (Clause 3)

Facts (already unit clauses, atomic literals are trivially in CNF):

CNF: SoilDry ... (Clause 4)

CNF: CropWheat ... (Clause 5)

(b) Resolution Refutation — Prove 'ApplyDripMethod'

Negate the goal and add it to the CNF knowledge base: $\neg \text{ApplyDripMethod}$... (Clause 6)

Full CNF set: {1: $\neg \text{SoilDry} \vee \text{IrrigationNeeded}$, 2: $\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$, 3: $\text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$, 4: SoilDry , 5: CropWheat , 6: $\neg \text{ApplyDripMethod}$ }

Resolution proof tree (each step resolves two clauses on a complementary literal):

Step 1: Resolve Clause 4 (SoilDry) with Clause 1 ($\neg \text{SoilDry} \vee \text{IrrigationNeeded}$)

→ IrrigationNeeded ... (Clause 7)

Step 2: Resolve Clause 7 (IrrigationNeeded) with Clause 2

($\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$)

→ $\neg \text{CropWheat} \vee \text{ApplyDripMethod}$... (Clause 8)

Step 3: Resolve Clause 5 (CropWheat) with Clause 8 ($\neg \text{CropWheat} \vee \text{ApplyDripMethod}$)

→ ApplyDripMethod ... (Clause 9)

Step 4: Resolve Clause 9 (ApplyDripMethod) with Clause 6 ($\neg \text{ApplyDripMethod}$)

→ $\square$ (empty clause)

Proof tree (indented view):

$\square$

/ \

ApplyDripMethod(9) $\neg \text{ApplyDripMethod}(6)$

/ \

CropWheat(5) $\neg \text{CropWheat} \vee \text{ApplyDripMethod}(8)$

/ \

IrrigationNeeded(7) $\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}(2)$

/ \

SoilDry(4) $\neg \text{SoilDry} \vee \text{IrrigationNeeded}(1)$

Since resolving the negated goal with the KB derives the empty clause $\square$ (a contradiction), the theorem ApplyDripMethod is proved TRUE by refutation.

(c) Effect of Removing the Fact 'SoilDry'

If SoilDry is removed from the KB, the fact clause (Clause 4: SoilDry) is no longer available, so Step 1 of the resolution proof above cannot be performed.

- Clause 1 ($\neg \text{SoilDry} \vee \text{IrrigationNeeded}$) can no longer be resolved into IrrigationNeeded, because there is no unit clause SoilDry to cancel $\neg \text{SoilDry}$.
- Consequently, IrrigationNeeded is no longer derivable, so Clause 2 cannot fire, and ApplyDripMethod can no longer be proved true. Modus Ponens simply has no premise to apply — the chain of forward inference is broken at the very first rule.
- Whether CropAtRisk follows depends on the reasoning pattern used:
 - Under the Closed-World Assumption (CWA), commonly used in rule-based expert systems: since ApplyDripMethod cannot be proved true, it is treated as False ($\neg \text{ApplyDripMethod}$). Combined with the fact CropWheat = True, Rule P3 ($\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$) fires by Modus Ponens, and CropAtRisk IS derived.
 - Under classical (open-world) first-order/propositional logic: failure to prove ApplyDripMethod does NOT logically entail $\neg \text{ApplyDripMethod}$ (this is the Negation-as-Failure fallacy). In that case, ApplyDripMethod remains simply UNKNOWN, so P3's premise $\neg \text{ApplyDripMethod}$ is also unproven, and CropAtRisk CANNOT be soundly derived either.

Conclusion: Removing 'SoilDry' breaks the forward chain at the earliest rule. Whether 'CropAtRisk' still follows is not absolute — it depends on whether the agent reasons under the Closed-World Assumption (then CropAtRisk = True) or under strict classical entailment (then CropAtRisk remains undetermined).

Case Study 4 — Wumpus World Knowledge Agent

(a) Belief State from Percepts — Modus Ponens Derivations

Percepts received: Stench[1,2] = True, Breeze[1,1] = True, Glitter[2,2] = True

Step	Premise (Fact)	Rule Applied (Modus Ponens: $P, P \rightarrow Q \vdash Q$)	Conclusion Derived
1	Stench[1,2] = True	Rule 1: Stench[1,2] $\rightarrow$ Wumpus adjacent to [1,2]	Wumpus is adjacent to [1,2]
2	Breeze[1,1] = True	Rule 2: Breeze[1,1] $\rightarrow$ Pit adjacent to [1,1]	Pit is adjacent to [1,1]
3	Glitter[2,2] = True	Rule 3: Glitter[x,y] $\rightarrow$ Gold at [x,y], x=2,y=2	Gold is at [2,2]
4	Agent occupied [1,1] earlier and survived (no pit fall)	Rule 4: $\neg$ Wumpus[x,y] $\wedge$ $\neg$ Pit[x,y] $\rightarrow$ Safe[x,y]	Safe[1,1] confirmed
5	Agent occupied [1,2] and survived (no Wumpus attack)	Rule 4 applied at [1,2]	Safe[1,2] confirmed

All derived conclusions: {Wumpus adjacent to [1,2]; Pit adjacent to [1,1]; Gold at [2,2]; Safe[1,1]; Safe[1,2]}.

(b) Forward Chaining — Possible Wumpus and Pit Locations

Grid adjacency used: cells adjacent to [1,2] are {[1,1], [1,3], [2,2]}; cells adjacent to [1,1] are {[1,2], [2,1]}.

Step 1: Stench[1,2]=True --(Rule 1)--> Wumpus $\in$ {[1,1], [1,3], [2,2]}

Step 2: [1,1] already visited and confirmed Safe (Step a-4)

--(elimination)--> Wumpus $\in$ {[1,3], [2,2]}

Step 3: Breeze[1,1]=True --(Rule 2)--> Pit $\in$ {[1,2], [2,1]}

Step 4: [1,2] already visited and confirmed Safe (Step a-5)

--(elimination)--> Pit $\in$ {[2,1]}

Step 5: Glitter[2,2]=True $\rightarrow$ (Rule 3) $\rightarrow$ Gold[2,2] = True

Result of forward chaining:

- Wumpus is most likely located at [1,3] or [2,2] — its exact cell is not yet disambiguated with only these three percepts.
- Pit is located at [2,1] (the only remaining unvisited cell adjacent to the breeze source [1,1]).
- Gold is confirmed at [2,2].

(c) Safety Evaluation of the Path [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2]

Cell	Status from KB	Safe[x,y] via Rule 4?	Justification
[1,1]	Visited, agent survived	Yes — Safe[1,1] proved	$\neg$ Wumpus[1,1] $\wedge$ $\neg$ Pit[1,1] confirmed empirically by successful occupation
[1,2]	Visited, agent survived	Yes — Safe[1,2] proved	$\neg$ Wumpus[1,2] $\wedge$ $\neg$ Pit[1,2] confirmed empirically by successful occupation
[2,2]	Unvisited; is a candidate Wumpus cell from part (b)	No — cannot be proved	$\neg$ Wumpus[2,2] is NOT established (Wumpus $\in$ {[1,3],[2,2]}); Rule 4's premise is unsatisfied, so Safe[2,2] does not follow

Conclusion: The path [1,1] $\rightarrow$ [1,2] is logically SAFE, since both cells have been empirically confirmed safe using Rule 4. However, the final step [1,2] $\rightarrow$ [2,2] is NOT provably safe: [2,2] remains a candidate Wumpus location from the forward-chaining result in part (b), and Rule 4 cannot be applied because $\neg$ Wumpus[2,2] has not been established. Therefore, the overall path to collect the gold at [2,2] cannot be certified as safe using the current knowledge base.

To make the path safe, the agent would need additional disambiguating percepts (e.g., checking for stench at [1,3] or [2,1] from an already-safe adjacent cell) to eliminate one of the two remaining Wumpus

candidates before entering [2,2], or it must accept the calculated risk based on probabilistic reasoning if forced to choose without further information.